

Computing Publications with Major Personal Contributions

Oliver Gutsche

May 10, 2026

1. Building an AI-native Research Ecosystem for Experimental Particle Physics: A Community Vision

Aarrestad, Thea Klæboe et al.

[arXiv:2602.17582](#) | 2026-02-19

2. Interpretable Models for Workflow Differentiation in High-Performance Scientific Networks

Giannakou, Anna et al.

(2026) 349 | [DOI:10.1109/ICNC68183.2026.11416957](#) | 2026

3. Evaluating Application Characteristics for GPU Portability Layer Selection

Atif, Mohammad et al.

[arXiv:2601.17526](#) | 2026-01-24

4. Application of performance portability solutions for GPUs and many-core CPUs to track reconstruction kernels

Kwok, Ka Hei Martin et al.

[EPJ Web Conf. 295 \(2024\) 11003](#) | [arXiv:2401.14221](#) | [DOI:10.1051/epjconf/202429511003](#) | 2024-01-25

5. The U.S. CMS HL-LHC R D Strategic Plan

Gutsche, Oliver et al.

[EPJ Web Conf. 295 \(2024\) 04050](#) | [arXiv:2312.00772](#) | [DOI:10.1051/epjconf/202429504050](#) | 2023-12-01

6. A Ceph S3 Object Data Store for HEP

Smith, Nick et al.

[EPJ Web Conf. 295 \(2024\) 01003](#) | [arXiv:2311.16321](#) | [DOI:10.1051/epjconf/202429501003](#) | 2023-11-27

7. Evaluating Portable Parallelization Strategies for Heterogeneous Architectures in High Energy Physics

Atif, Mohammad et al.

[arXiv:2306.15869](#) | 2023-06-27

8. Detector R D needs for the next generation e^+e^- collider

Apresyan, A. et al.

[arXiv:2306.13567](#) | 2023-06-23

9. Automated Network Services for Exascale Data Movement

Balcas, Justas et al.

[EPJ Web Conf. 295 \(2024\) 01009](#) | [DOI:10.1051/epjconf/202429501009](#) | 2024

10. IRIS-HEP Strategic Plan for the Next Phase of Software Upgrades for HL-LHC Physics

Bockelman, Brian et al.

[arXiv:2302.01317](#) | 2023-02-02

11. The Future of High Energy Physics Software and Computing

Elvira, V. Daniel et al.

[arXiv:2210.05822](#) | [DOI:10.2172/1898754](#) | 2022-10-11

12. Snowmass Computational Frontier: Topical Group Report on Experimental Algorithm Parallelization

Cerati, G. et al.

[arXiv:2209.07356](#) | 2022-09-15

13. Portability: A Necessary Approach for Future Scientific Software

Bhattacharya, Meghna et al.

[arXiv:2203.09945](#) | 2022-03-15

14. Coffea: Columnar Object Framework For Effective Analysis

Smith, Nicholas et al.

[EPJ Web Conf. 245 \(2020\) 06012](#) | [arXiv:2008.12712](#) | [DOI:10.1051/epjconf/202024506012](#) | 2020-08-28

15. Response to NITRD, NCO, NSF Request for Information on "Update to the 2016 National Artificial Intelligence Research and Development Strategic Plan"

Amundson, J. et al.

[arXiv:1911.05796](#) | [DOI:10.2172/1592156](#) | 2019-11-04

16. Striped Data Analysis Framework

Gutsche, Oliver et al.

[EPJ Web Conf. 245 \(2020\) 06042](#) | [DOI:10.1051/epjconf/202024506042](#) | 2020

17. CMS Computing Resources: Meeting the Demands of the High-Luminosity LHC Physics Program

Lange, David et al.

[EPJ Web Conf. 214 \(2019\) 03055](#) | [DOI:10.1051/epjconf/201921403055](#) | 2019-09-17

18. New Technologies for Discovery

Ahmed, Z. et al.

[arXiv:1908.00194](#) | 2019-07-31

19. Using Big Data Technologies for HEP Analysis

Cremonesi, Matteo et al.

[EPJ Web Conf. 214 \(2019\) 06030](#) | [arXiv:1901.07143](#) | [DOI:10.1051/epjconf/201921406030](#) | 2019-01-21

20. HEP Software Foundation Community White Paper Working Group – Data Organization, Management and Access (DOMA)

Berzano, Dario et al.

[arXiv:1812.00761](#) | 2018-11-30

21. HEP Software Foundation Community White Paper Working Group – Visualization

Bellis, Matthew et al.

[arXiv:1811.10309](#) | 2018-11-26

22. HPC resource integration into CMS Computing via HEPCloud

Hufnagel, Dirk et al.

[EPJ Web Conf. 214 \(2019\) 03031](#) | [DOI:10.1051/epjconf/201921403031](#) | 2018-10-16

23. HEP Software Foundation Community White Paper Working Group - Data Analysis and Interpretation

Bauerdick, Lothar et al.

[arXiv:1804.03983](#) | 2018-04-09

24. A Roadmap for HEP Software and Computing R D for the 2020s

Albrecht, Johannes et al.

[Comput.Softw.Big Sci. 3 \(2019\) 7](#) | [arXiv:1712.06982](#) | [DOI:10.1007/s41781-018-0018-8](#) | 2017-12-18

25. Experience in using commercial clouds in CMS

Bauerdick, L. et al.

[J.Phys.Conf.Ser. 898 \(2017\) 052019](#) | [DOI:10.1088/1742-6596/898/5/052019](#) | 2017-11-22

26. CMS Analysis and Data Reduction with Apache Spark

Gutsche, Oliver et al.

[J.Phys.Conf.Ser. 1085 \(2018\) 042030](#) | [arXiv:1711.00375](#) | [DOI:10.1088/1742-6596/1085/4/042030](#) | 2017-10-31

27. HEPCloud, a New Paradigm for HEP Facilities: CMS Amazon Web Services Investigation

Holzman, Burt et al.

[Comput.Softw.Big Sci. 1 \(2017\) 1](#) | [arXiv:1710.00100](#) | [DOI:10.1007/s41781-017-0001-9](#) | 2017-09-29

28. Striped Data Server for Scalable Parallel Data Analysis

Chang, Jin et al.

[J.Phys.Conf.Ser. 1085 \(2018\) 042035](#) | [DOI:10.1088/1742-6596/1085/4/042035](#) | 2017

29. Big Data in HEP: A comprehensive use case study

Gutsche, Oliver et al.

[J.Phys.Conf.Ser. 898 \(2017\) 072012](#) | [arXiv:1703.04171](#) | [DOI:10.1088/1742-6596/898/7/072012](#) | 2017-03-12

30. Dark Matter and Super Symmetry: Exploring and Explaining the Universe with Simulations at the LHC

Gutsche, Oliver et al.

[\(2017\) 4](#) | [DOI:10.1109/WSC.2016.7822075](#) | 2017-01-19

31. ASCR/HEP Exascale Requirements Review Report

Habib, Salman et al.

[arXiv:1603.09303](#) | 2016-03-30

32. Pooling the resources of the CMS Tier-1 sites

Apyan, A. et al.

[J.Phys.Conf.Ser. 664 \(2015\) 042056](#) | [DOI:10.1088/1742-6596/664/4/042056](#) | 2015-12-23

33. Using the GlideinWMS System as a Common Resource Provisioning Layer in CMS

Balcas, J. et al.

[J.Phys.Conf.Ser. 664 \(2015\) 062031](#) | [DOI:10.1088/1742-6596/664/6/062031](#) | 2015-12-23

34. Pushing HTCondor and glideinWMS to 200K+ Jobs in a Global Pool for CMS before Run 2

Balcas, J. et al.

[J.Phys.Conf.Ser. 664 \(2015\) 062030](#) | [DOI:10.1088/1742-6596/664/6/062030](#) | 2015-12-23

35. Diversity in Computing Technologies and Strategies for Dynamic Resource Allocation

Garzoglio, G. et al.

[J.Phys.Conf.Ser. 664 \(2015\) 012001](#) | [DOI:10.1088/1742-6596/664/1/012001](#) | 2015-12-23

36. Fermilab Computing at the Intensity Frontier

Group, Craig et al.

[J.Phys.Conf.Ser. 664 \(2015\) 032012](#) | [DOI:10.1088/1742-6596/664/3/032012](#) | 2015-12-23

37. High Energy Physics Forum for Computational Excellence: Working Group Reports (I. Applications Software II. Software Libraries and Tools III. Systems)

Habib, Salman et al.

[arXiv:1510.08545](#) | 2015-10-28

38. CMS Computing Operations During Run 1

Adelman, J. et al.

[J.Phys.Conf.Ser. 513 \(2014\) 032040](#) | [DOI:10.1088/1742-6596/513/3/032040](#) | 2013-10-30

39. Evolution of the Pilot Infrastructure of CMS: Towards a Single GlideinWMS Pool

Belforte, S. et al.

[J.Phys.Conf.Ser. 513 \(2014\) 032041](#) | [DOI:10.1088/1742-6596/513/3/032041](#) | 2014

40. Deployment of a WLCG network monitoring infrastructure based on the perfSONAR-PS technology

Campana, S. et al.

[J.Phys.Conf.Ser. 513 \(2014\) 062008](#) | [DOI:10.1088/1742-6596/513/6/062008](#) | 2014

41. Opportunistic Resource Usage in CMS

Kreuzer, Peter et al.

[J.Phys.Conf.Ser. 513 \(2014\) 062028](#) | [DOI:10.1088/1742-6596/513/6/062028](#) | 2014

42. Towards a Centralized Grid Speedometer

Dzhunov, I. et al.

[J.Phys.Conf.Ser. 513 \(2014\) 032028](#) | [DOI:10.1088/1742-6596/513/3/032028](#) | 2014

43. CMS experience of running glideinWMS in High Availability mode

Sfiligoi, I. et al.

[J.Phys.Conf.Ser. 513 \(2014\) 032086](#) | [DOI:10.1088/1742-6596/513/3/032086](#) | 2014

44. A New Era for Central Processing and Production in CMS

Fajardo, E. et al.

[J.Phys.Conf.Ser. 396 \(2012\) 042018](#) | [DOI:10.1088/1742-6596/396/4/042018](#) | 2012-06-27

45. No File Left Behind - Monitoring Transfer Latencies in PhEDEx

Chwalek, T. et al.

[J.Phys.Conf.Ser. 396 \(2012\) 032089](#) | [DOI:10.1088/1742-6596/396/3/032089](#) | 2012

46. CMS Data Transfer Operations After the First Years of LHC Collisions

Kaselis, Rapolas et al.

[J.Phys.Conf.Ser. 396 \(2012\) 042033](#) | [DOI:10.1088/1742-6596/396/4/042033](#) | 2012

47. Monitoring Techniques and Alarm Procedures for CMS Services and Sites in WLCG

Molina-Perez, J. et al.

[J.Phys.Conf.Ser. 396 \(2012\) 042041](#) | [DOI:10.1088/1742-6596/396/4/042041](#) | 2012

48. CMS distributed computing workflow experience

Adelman-McCarthy, Jennifer et al.

[J.Phys.Conf.Ser. 331 \(2011\) 072019](#) | [DOI:10.1088/1742-6596/331/7/072019](#) | 2011

49. Experience Building and Operating the CMS Tier-1 Computing Centres

Albert, M. et al.

[J.Phys.Conf.Ser. 219 \(2010\) 072035](#) | [DOI:10.1088/1742-6596/219/7/072035](#) | 2010

50. Use of glide-ins in CMS for production and analysis

Bradley, D. et al.

[J.Phys.Conf.Ser. 219 \(2010\) 072013](#) | [DOI:10.1088/1742-6596/219/7/072013](#) | 2010

51. Validation of Software Releases for CMS

Gutsche, Oliver et al.

[J.Phys.Conf.Ser. 219 \(2010\) 042040](#) | [DOI:10.1088/1742-6596/219/4/042040](#) | 2009-03

52. Stand-alone Cosmic Muon Reconstruction Before Installation of the CMS Silicon Strip Tracker

Adam, W. et al.

[JINST 4 \(2009\) P05004](#) | [arXiv:0902.1860](#) | [DOI:10.1088/1748-0221/4/05/P05004](#) | 2009-02

53. Large scale job management and experience in recent data challenges within the LHC CMS experiment

Evans, D. et al.

[PoS ACAT08 \(2008\) 032](#) | [DOI:10.22323/1.070.0032](#) | 2008

54. The CMS Remote Analysis Builder (CRAB)

Spiga, D. et al.

[Lect.Notes Comput.Sci. 4873 \(2007\) 580](#) | [DOI:10.1007/978-3-540-77220-0_52](#) | 2007

55. WLCG scale testing during CMS data challenges

Gutsche, Oliver et al.

[J.Phys.Conf.Ser. 119 \(2008\) 062033](#) | [DOI:10.1088/1742-6596/119/6/062033](#) | 2008

56. CRAB: The CMS distributed analysis tool development and design

Spiga, D. et al.

[Nucl.Phys.B Proc.Suppl. 177-178 \(2008\) 267](#) | [DOI:10.1016/j.nuclphysbps.2007.11.124](#) | 2008

57. Status and evolution of CRAB

Farina, Fabio et al.

[PoS ACAT \(2007\) 020](#) | [DOI:10.22323/1.050.0020](#) | 2007

58. A ROOT based client server event display for the ZEUS experiment

Kind, O. et al.

[eConf C0303241 \(2003\) MOLT002](#) | [arXiv:hep-ex/0305095](#) | 2003-05